

2501-PYL1003 (Mechanics & STR) — Practice Mid-Sem Paper 2

Full Marks: 50 | Time: 2 hours | Instructor Style: Pradipta Ghosh (IIT Delhi)

Instructions

An MCQ can also be an MSQ (multiple correct options). No credit without justification/derivation. Show all intermediate steps. Numerical answers without units lose marks. Approximations must be justified explicitly.

Q1. The energy density of blackbody radiation is proposed to depend on the temperature T , the speed of light c , and the Boltzmann constant k_B (dimension $ML^2T^{-2}\Theta^{-1}$, where Θ is the dimension of temperature) as

$$u = AT^a c^b k_B^c$$

where u has dimensions of energy per unit volume and A is dimensionless.

(i) Using dimensional analysis, determine a, b, c uniquely (you will need to also introduce Planck's constant h dimensionally if you find the system under-determined — explain clearly why or why not this is necessary here). **(4)**

(ii) Comment on whether dimensional analysis alone could ever fix the dimensionless constant A , and explain in one or two sentences what additional physics would be needed. **(2)**

Q2 (MSQ). Consider the quantity $Y = \frac{q^2}{4\pi\epsilon_0\hbar c}$ (the fine structure constant), where q is electric charge (dimension IT , with I = current), ϵ_0 is permittivity of free space, $\hbar$ is the reduced Planck constant, and c is the speed of light. Which of the following is/are correct? **(1+1+1 = 3)**

a. Y is dimensionless. b. Y has the same dimension as angular momentum. c. Since Y is dimensionless, its numerical value cannot depend on the choice of unit system (SI vs. CGS). d. Y has the dimension $M^0L^0T^0$.

Q3. A particle starts from rest at the origin and moves along the x -axis with velocity-dependent acceleration $a = g - kv^2$ (this models fall through a resistive medium, with $g, k > 0$ constants, $v \geq 0$ taken downward as positive).

(i) Find the terminal velocity v_T of the particle (i.e., v as $t \rightarrow \infty$), without solving the full differential equation. **(1)**

(ii) Solve the differential equation exactly to find $v(t)$. **(4)**

(iii) Show that your answer to (ii) correctly reduces to $v = gt$ for $t \ll 1/\sqrt{gk}$ (i.e., recovers free-fall at short times). **(2)**

Q4. A uniform chain of total mass M and length L lies coiled in a heap on a table. One end is pulled straight up with a constant **velocity** v_0 (not constant force). Let $x(t)$ be the length of chain already lifted off the table at time t , so $x = v_0 t$.

(i) Using the variable-mass form of Newton's second law appropriate for a chain picking up stationary mass from rest, find the force $F(t)$ that must be applied by the hand at time t , in terms of M , L , v_0 , g , and t (valid while $x < L$). **(4)**

(ii) Explain physically why $F(t)$ contains a term that does **not** vanish even though the acceleration of the moving part of the chain is zero (since v_0 is constant). **(2)**

Q5. Two blocks A (mass m) and B (mass $2m$) are connected by a light inextensible string over a frictionless, massless pulley fixed at the edge of a rough horizontal table. Block A sits on the table (coefficient of kinetic friction μ with the table), and block B hangs vertically. The system is released from rest.

(i) Derive the acceleration of the system in terms of m , μ , g , clearly stating the condition on μ for the system to accelerate at all. **(4)**

(ii) Suppose instead the table itself (along with the pulley fixed to it) is accelerated horizontally with constant acceleration a_0 in the direction from A toward the pulley. Re-derive the acceleration of block B relative to the table in this non-inertial frame, including the pseudo-force on A . **(4)**

Q6. A point P has cylindrical polar coordinates $\rho = 10$, $\phi = 143.13^\circ$, $z = -24$.

(i) Find the Cartesian coordinates of P . **(2)**

(ii) Find the spherical polar coordinates (r, θ, ϕ) of P , rounded to two decimal places. **(3)**

(iii) A vector at P is given as $\vec{B} = 5\hat{\rho} - 3\hat{\phi} + 2\hat{z}$ in cylindrical basis vectors. Express $\vec{B}$ in Cartesian components at P . **(3)**

Q7. Starting from the Cartesian expression $\vec{r} = r \sin \theta \cos \phi \hat{i} + r \sin \theta \sin \phi \hat{j} + r \cos \theta \hat{k}$:

(i) Derive $\hat{\phi}$ in terms of $\hat{i}$, $\hat{j}$ only, using the definition $\hat{u} = \frac{\partial \vec{r} / \partial u}{|\partial \vec{r} / \partial u|}$. **(2)**

(ii) By direct differentiation with respect to time of your Cartesian expressions for $\hat{\theta}$ and $\hat{\phi}$, **prove** that

$$\frac{d\hat{\phi}}{dt} = -\dot{\phi} \sin \theta \hat{r} - \dot{\phi} \cos \theta \hat{\theta}$$

Show every step; do not simply quote the result. **(5)**

Q8. A particle moves on the surface of a sphere of fixed radius $r = R$ (so $\dot{r} = 0$), with $\theta(t)$ and $\phi(t)$ both time-varying.

- (i) Using $\vec{v} = \dot{r} \hat{r} + r\dot{\theta} \hat{\theta} + r \sin \theta \dot{\phi} \hat{\phi}$, write down the speed $|\vec{v}|$ of the particle in terms of $R, \dot{\theta}, \dot{\phi}, \theta$. **(2)**
- (ii) If the particle moves along a curve of constant ϕ (a "meridian") with $\dot{\theta} = \omega_0 = \text{constant}$, find the magnitude of its acceleration $|\vec{a}|$, using $\vec{a} = (\ddot{r} - r\dot{\theta}^2 - r \sin^2 \theta \dot{\phi}^2) \hat{r} + (\dots) \hat{\theta} + (\dots) \hat{\phi}$ [you should look up / derive which terms survive]. **(3)**

Q9. For an inverse-square attractive central force $F(r) = -k/r^2$, the total mechanical energy is E and angular momentum is L , for a particle of reduced mass μ .

- (i) **Prove** that the eccentricity of the orbit is given by

$$\epsilon = \sqrt{1 + \frac{2EL^2}{\mu k^2}}$$

starting from the effective potential energy method (i.e., from $\frac{1}{2}\mu\dot{r}^2 + \frac{L^2}{2\mu r^2} - \frac{k}{r} = E$) and the substitution $u = 1/r$. **(6)**

- (ii) Using this result, state the sign condition on E that separates bound ($\epsilon < 1$) from unbound ($\epsilon \geq 1$) orbits, and explain why this makes physical sense. **(2)**

Q10 (MSQ). A satellite is in a circular orbit of radius r_0 around the Earth. At some instant, its speed is **suddenly** increased by firing a thruster, without changing its position or the direction of velocity (which remains tangential). Which of the following is/are correct about the new orbit? **(1+1+1+1=4)**

- a. The new orbit is an ellipse with the point of the speed boost as the perigee (point of closest approach).
b. The new orbit's semi-major axis is larger than r_0 . c. The angular momentum of the satellite about the Earth's center increases as a result of the boost. d. If the speed is increased sufficiently, the satellite can escape to infinity, and the minimum speed for this to occur (starting from radius r_0) is $\sqrt{2}$ times the original circular orbital speed.

Q11. A block of mass $M = 4 \text{ kg}$ hangs from a system of two movable pulleys and one fixed pulley (all massless, frictionless), arranged such that the free end of the string is held by a person and the

mechanical advantage of the system is exactly 4 (i.e., the person's applied tension supports $1/4$ of the total weight when in equilibrium).

(i) If the system is in equilibrium, find the tension T the person must apply, and the tension in each of the segments directly supporting the block. Draw the full pulley arrangement consistent with mechanical advantage 4 using **only** two movable and one fixed pulley. **(4)**

(ii) Now the person pulls with a **constant** force of $T = 12 \text{ N}$ (greater than the equilibrium value). Find the upward acceleration of the block M , and the corresponding downward acceleration of the person's hand, in terms of the given numbers ($g = 10 \text{ m/s}^2$). **(3)**

Q12. A particle of mass 1 kg moves in one dimension under a potential energy $U(x) = 4x^2 - x^4$ (SI units, valid for all real x).

(i) Find all equilibrium points and classify each as stable or unstable. **(3)**

(ii) If the particle is released from rest at $x = 0.5 \text{ m}$, determine (with justification, not by solving the ODE) whether its subsequent motion is bounded or unbounded. **(3)**

Q13. Three identical springs, each of natural length L_0 and spring constant k , connect two blocks of mass m each to two rigid walls and to each other, arranged in series along a frictionless horizontal line as: **wall — spring — block 1 — spring — block 2 — spring — wall**. Let x_1, x_2 be the displacements of blocks 1 and 2 from their respective equilibrium positions.

(i) Write down the coupled equations of motion for x_1 and x_2 . **(2)**

(ii) Find the normal mode frequencies of the system (i.e., the two values of ω for which $x_1, x_2 \propto \cos(\omega t)$ with fixed amplitude ratio). **(5)**

Q14. A rigid body consists of two point masses, m at $(1, 0, 0)$ and m at $(-1, 1, 0)$ (in meters), both measured from the origin O , at some instant.

(i) Compute the total angular momentum $\vec{L}$ of the system about O at this instant, if mass 1 has velocity $(0, 2, 0) \text{ m/s}$ and mass 2 has velocity $(0, 0, 3) \text{ m/s}$. **(3)**

(ii) A constant force $\vec{F} = (0, 0, 6m) \text{ N}$ is applied to mass 2 only (mass 1 is force-free). Compute the torque about O at this instant, and hence $\frac{d\vec{L}}{dt}$ at this instant. **(3)**

Q15. A 2 kg block sits on a rough incline of angle 37° ($\sin 37^\circ \approx 0.6$, $\cos 37^\circ \approx 0.8$), with $\mu_s = 0.5$ and $\mu_k = 0.4$. The block is initially held at rest and then released.

(i) Determine whether the block starts to slide, with justification. **(2)**

(ii) If it slides, find its acceleration down the incline, and the time taken to travel 4 m down the incline from rest, using $g = 10 \text{ m/s}^2$. **(3)**

(End of Paper 2 — 50 marks)